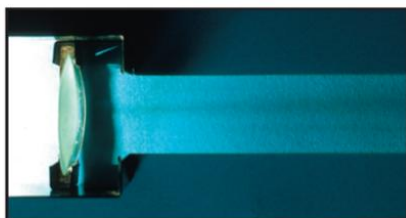
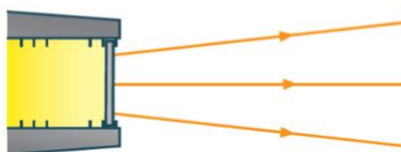
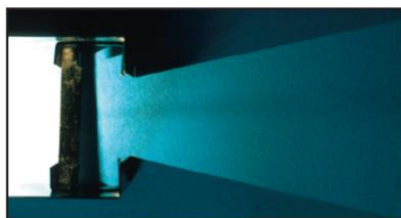


Light rays and light beams

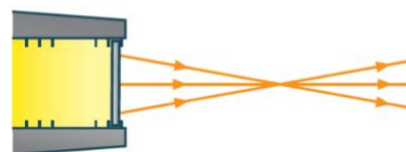
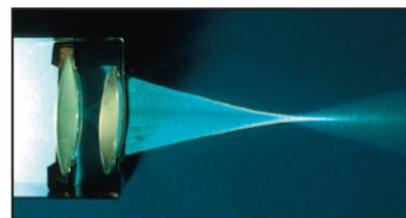
(Parallel beam)



(Divergent beam)



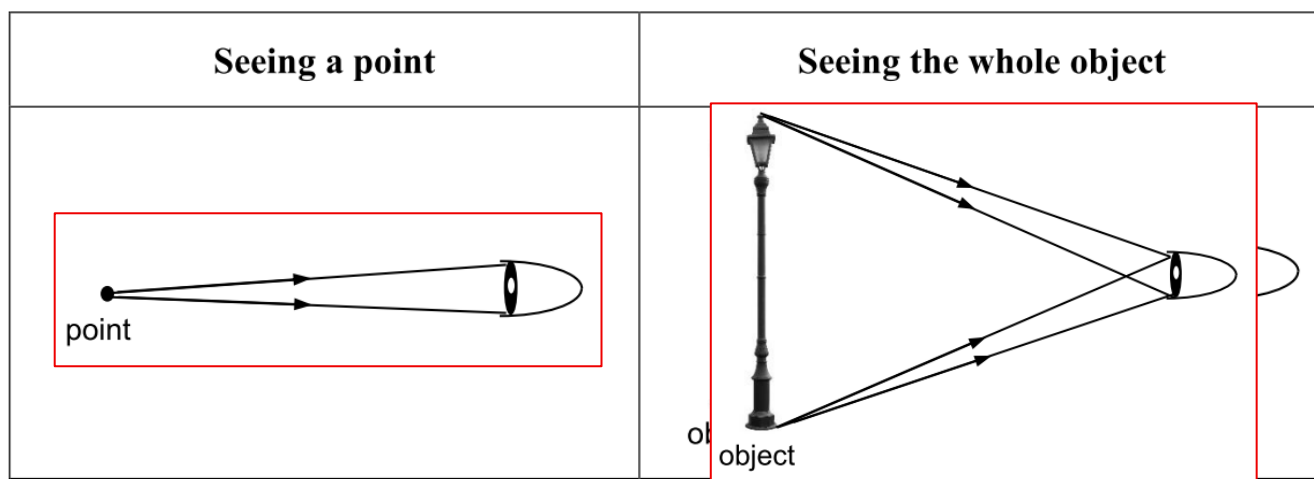
(Convergent beam)



- Light consists of **rays** travelling in straight lines.
- Each ray is represented by a solid straight line with an **arrow** showing the travelling direction.

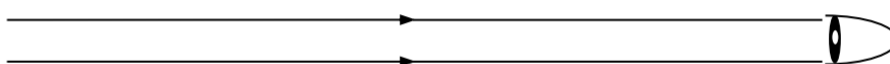
Drawing rays

- The light from an object **entering our eye** produces vision.
- The light from a point on a near object reaches our eyes as divergent rays. If the object moves away from us, the rays from the same point will become less divergent.



- The rays from a point on a distant object are regarded as **parallel**

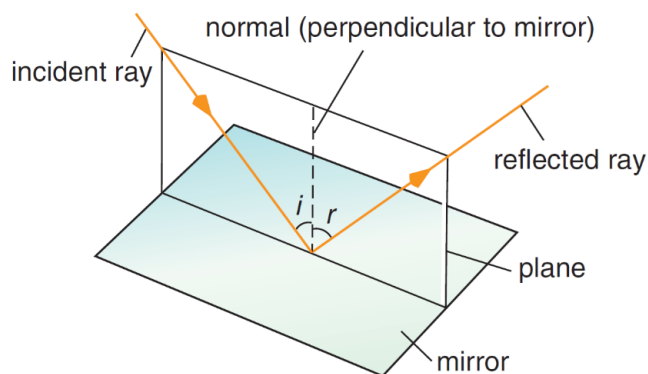
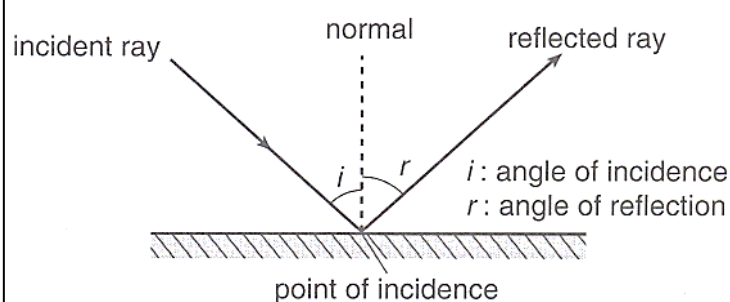
rays from a point on
a distant object



Reflection of light

Law of reflection:

- Angle of incidence (i) = Angle of reflection (r)
- The normal, incident ray and reflected ray all lie on the same plane.



- **Incident ray:** The light ray that approaches the mirror
- **Reflected ray:** The light ray that leaves the mirror after reflection
- **Normal:** An imaginary line perpendicular to the plane mirror at the point where the incident ray hits the mirror, represented by a dotted line
- **Angle of incidence:** The angle between the incident ray and the normal
- **Angle of reflection:** The angle between the reflected ray and the normal

Remarks: Light follows the same path if the direction of light is reversed. This is called the **reversibility of light**.

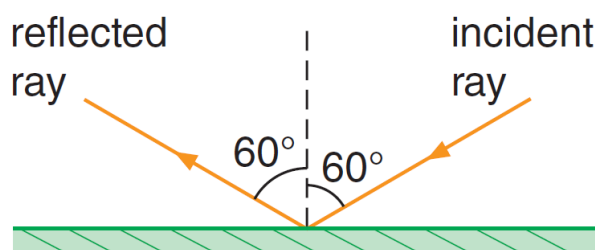
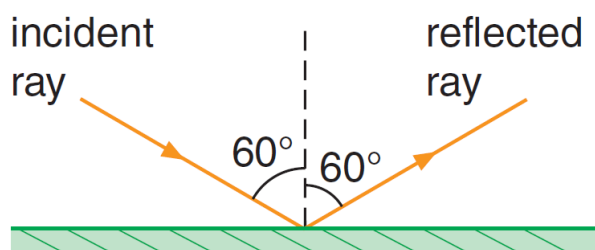
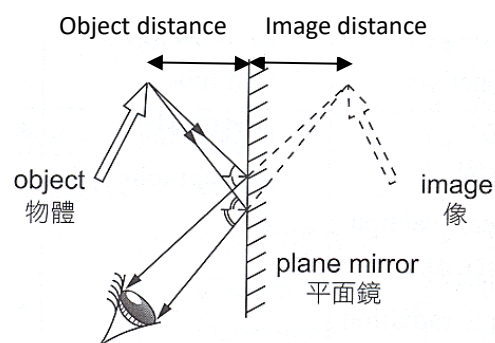
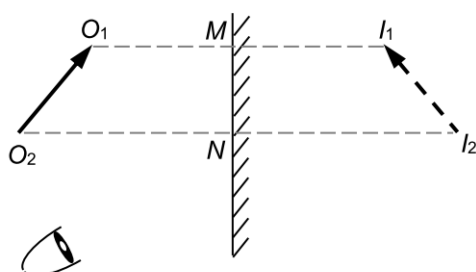


Image formed by plane mirror

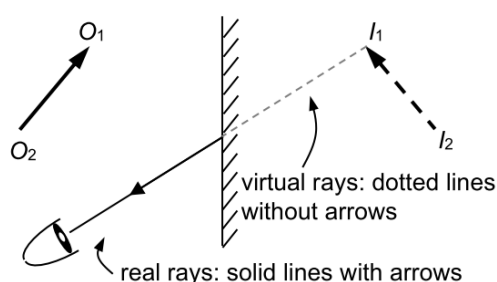
- Laterally inverted
- Same size as the object
- Erect
- Virtual
- Object distance = Image distance



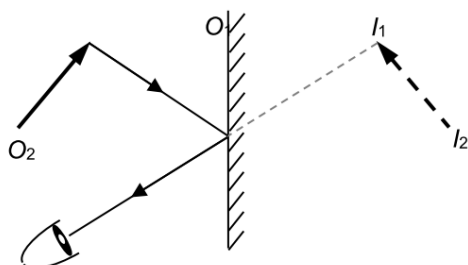
① Locate and construct the image I_1I_2 .



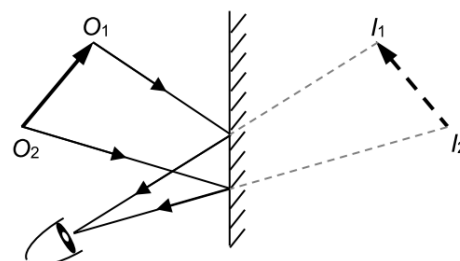
② Draw a ray from I_1 to the observer.



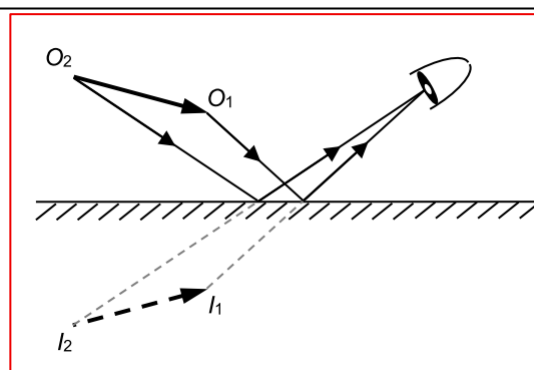
③ Draw a ray from O_1 to the points where the rays from I_1 meet the mirror.



④ Repeat steps ② and ③ for I_2 .



- A ray diagram can be drawn to show how we see an image formed by a plane mirror. Follow the steps and finish the ray diagram
- If the line joining an image point and the observer does not pass through the mirror, the observer cannot see the image.



Application of plane mirror:

- Dressing mirror
- Rear view driving mirror

Advantage of using plane mirror:

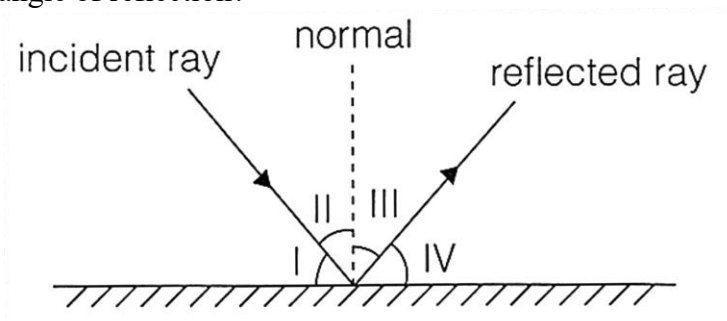
- Plane mirror gives a same size image

Disadvantage of using a plane mirror:

- Form multiple images

1. Which of the angles shown below is the angle of reflection?

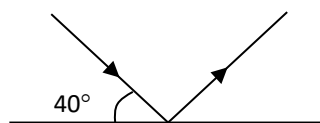
A. I
B. II
C. III
D. IV



C

2. A ray of light hits a plane mirror as shown in the figure. What is the angle of reflection?

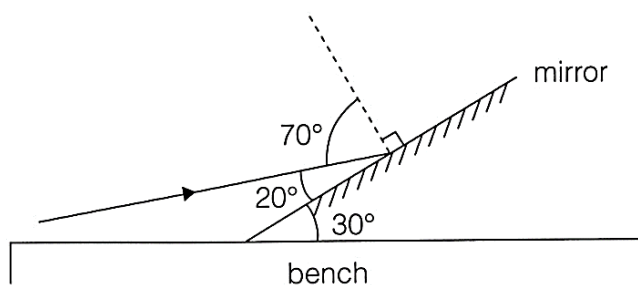
A 40°
B 50°
C 80°
D 100°



B

3. A mirror is tilted at an angle of 30° to the bench. A ray of light is directed so that it hits the mirror at an angle of 20° to the surface of the mirror. What is the angle of reflection?

A. 20°
B. 30°
C. 50°
D. 70°



D

4. In the figure below, what is the angle of reflection of the ray after being reflected by mirror M_2 ?

A 40°
B 45°
C 50°
D 55°

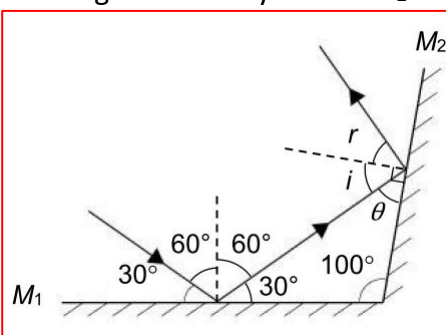
$$\theta + 30^\circ + 100^\circ = 180^\circ \quad (\angle \text{sum of } \Delta)$$

$$\theta = 50^\circ$$

$$i + \theta = 90^\circ$$

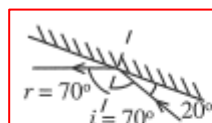
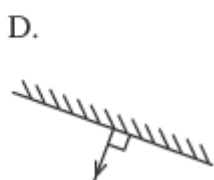
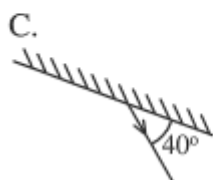
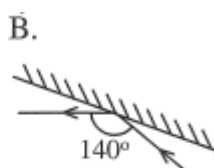
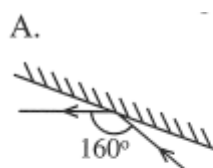
$$i = 40^\circ$$

By the laws of reflection, $r = i = 40^\circ$
The angle of reflection is 40° .



A

5. As shown below, a light ray strikes at an inclined plane mirror. Does which of the following ray diagrams correctly show the reflected light ray?



Incident angle $i = 90^\circ - 20^\circ = 70^\circ$
Reflected angle $r = 70^\circ$
The angle subtended by the reflected ray and the incident ray = $70^\circ + 70^\circ = 140^\circ$
The answer is B.

B

6. Which of the following is **not** the nature of an image formed by a plane mirror?

- A. Laterally inverted
- B. Image size equal to object size
- C. Real image
- D. Image distance equal to object distance

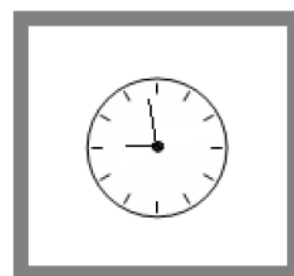
C The image formed by a plane mirror is virtual.

C

7. A clock is facing a plane mirror on a wall. The figure below shows the image of the clock in the mirror. What is the actual time at this instant?

- A. 2:58
- B. 3:02
- C. 8:58
- D. 12:13

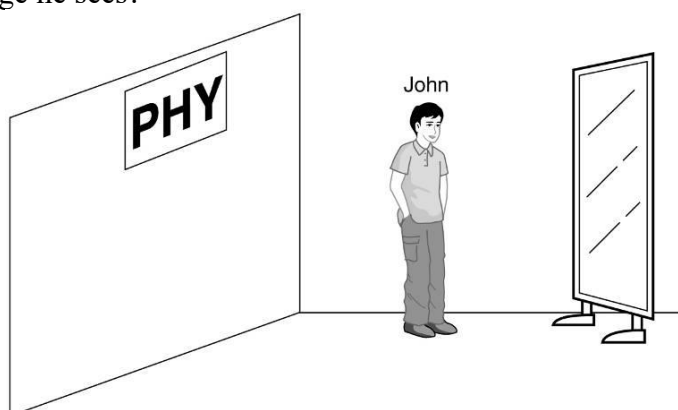
An image formed by a plane mirror is laterally inverted.
The clock looks like this:



B

8. John stands in front of a plane mirror in which he sees the image of three letters 'PHY' posted on the wall behind him. Which of the following is the image he sees?

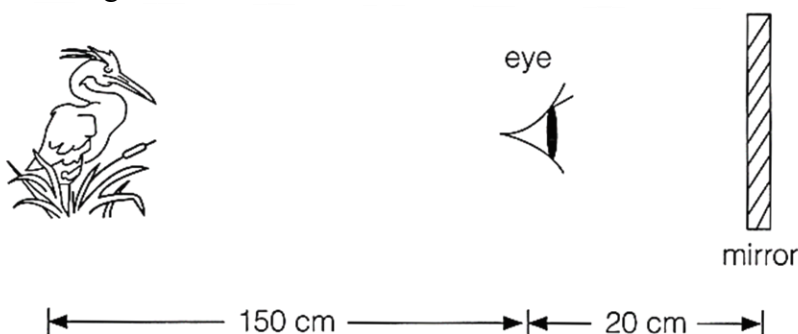
- A. PHY
- B. QHY
- C. YHP
- D. YH9



D

9. A lady is holding a mirror 20 cm in front of herself. She sees a bird that is 150 cm behind her. What is the distance from her eyes to the image of the bird?

- A. 150 cm
- B. 170 cm
- C. 190 cm
- D. 320 cm



C

10. A girl stands in front of a plane mirror. Her image is 4 m away from her. What is the distance between her and the mirror?

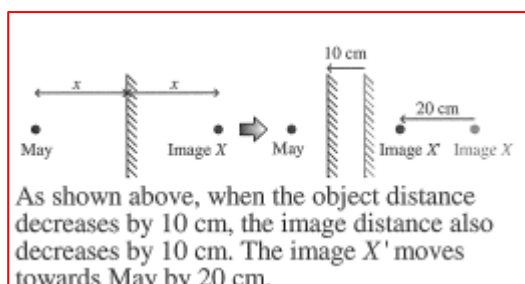
A. 2 m B. 3 m
C. 4 m C. 8 m

A Distance between girl and mirror = $\frac{4}{2} = 2 \text{ m}$

A

11. From the plane mirror held by May, she saw her face. If she moved the plane mirror 10 cm closer to herself, find the movement of the observed image of her face

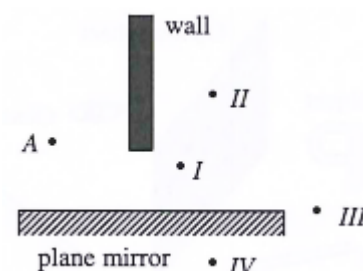
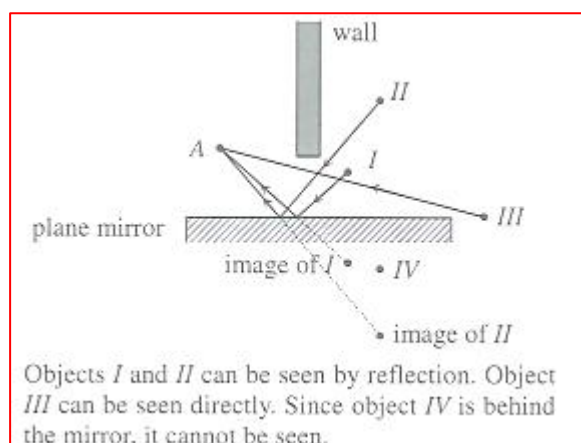
A. It moved towards May by 5 cm
B. It moved towards May by 10 cm
C. It moved towards May by 20 cm
D. It moved away from May by 20 cm



C

12. There are 4 point objects (denoted by I, II, III and IV) placed at the positions as shown. A man is standing at position A in front of a plane mirror. He can see objects

A. I and II only
B. II and III only
C. I, II and III only
D. II, III and IV only

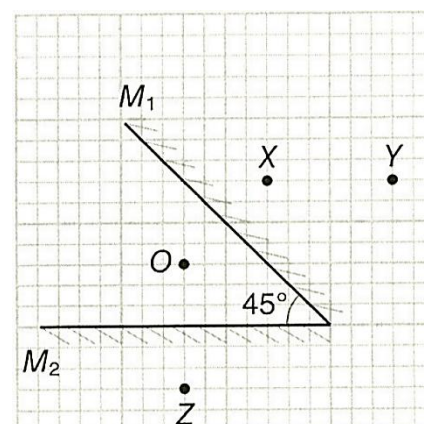


C

13. Two plane mirrors M_1 and M_2 are placed together such that the angle between them is 45° . An object O is placed between the mirrors as shown. Which of the following are the images formed by the mirrors?

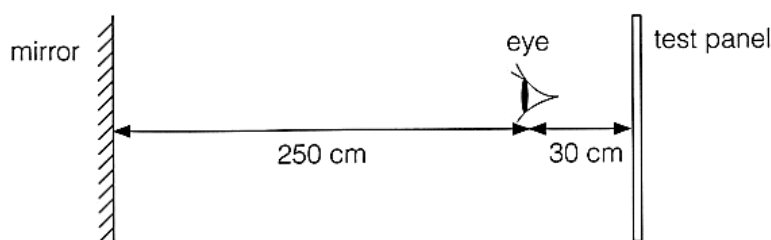
A. X and Y only
B. X and Z only
C. Y and Z only
D. X, Y and Z

D X is the image of O formed by M_1 .
Z is the image of O formed by M_2 .
Y is the image of Z formed by M_1 .



D

- 14 An optician's test panel is located at 30 cm behind the eyes of a patient who looks at a plane mirror 250 cm in front of him, as shown by the diagram below.



Calculate the actual distance of the image of the test panel from the patient's eyes.

530 cm

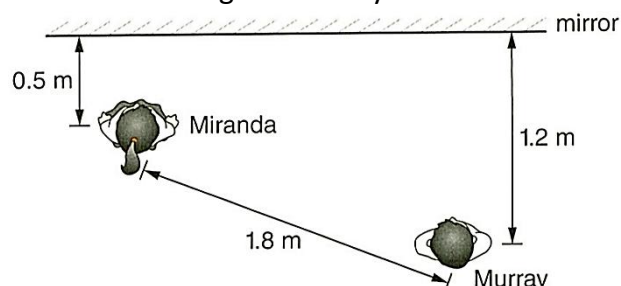
Image distance = _____

15. Miranda and Murray are 1.8 m apart. They stand in front of a mirror as shown.

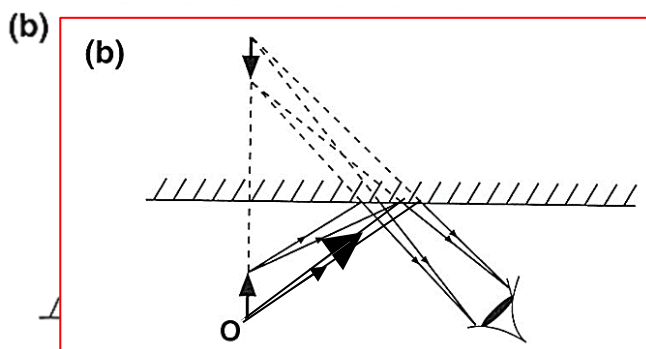
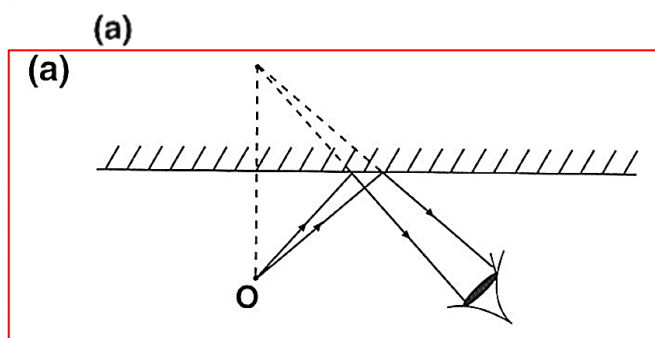
- (a) What is the distance between Miranda and her image?
(b) What is the distance between the image of Miranda and the image of Murray?

(a) Distance = $2 \times 0.5 = 1$ m

(b) 1.8 m

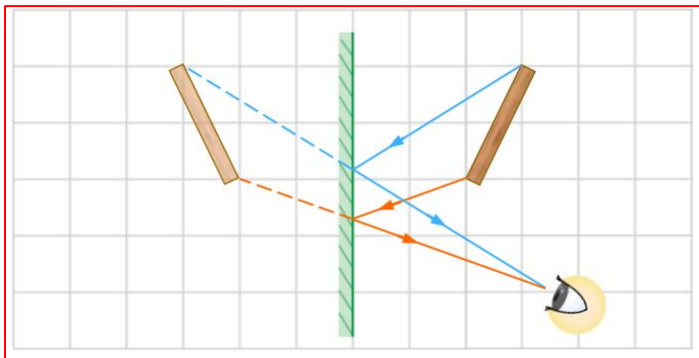


16. Draw a ray diagram with two light rays in each of the following to show how the eye sees the image of the object O.

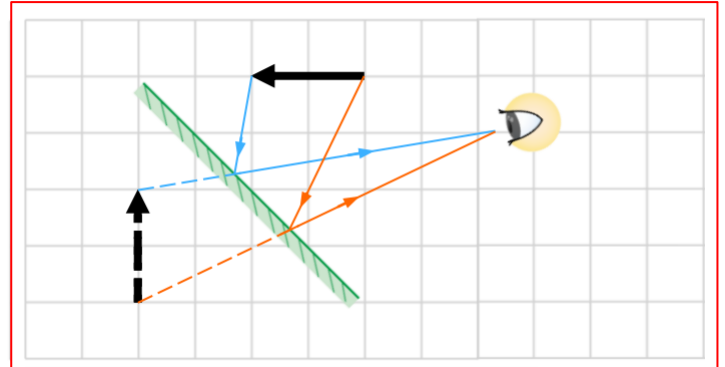


17. In each of the following cases, complete the ray diagram to show how the observer sees the whole image.

(a)

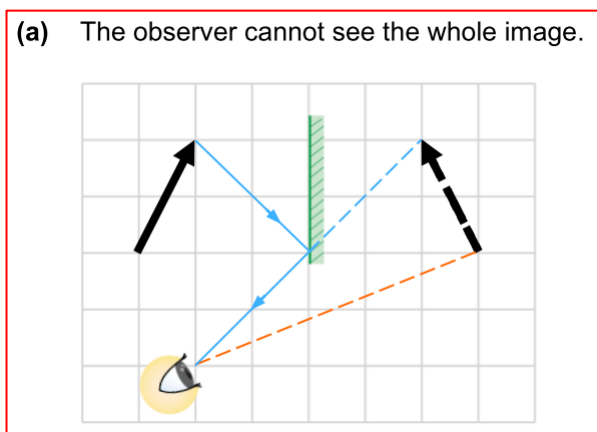


(b)

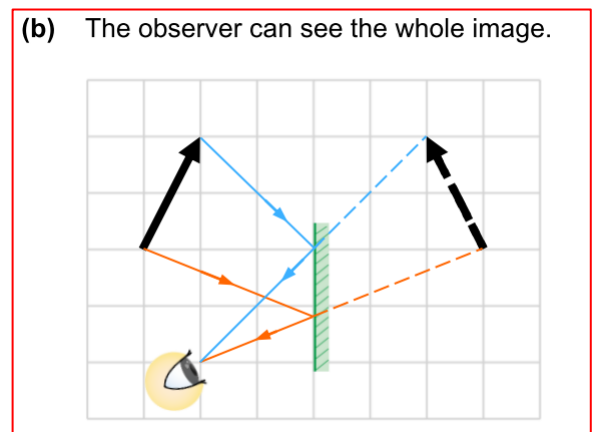


18. In each of the following cases, can the observer see the whole image? Draw ray diagrams to support your answers.

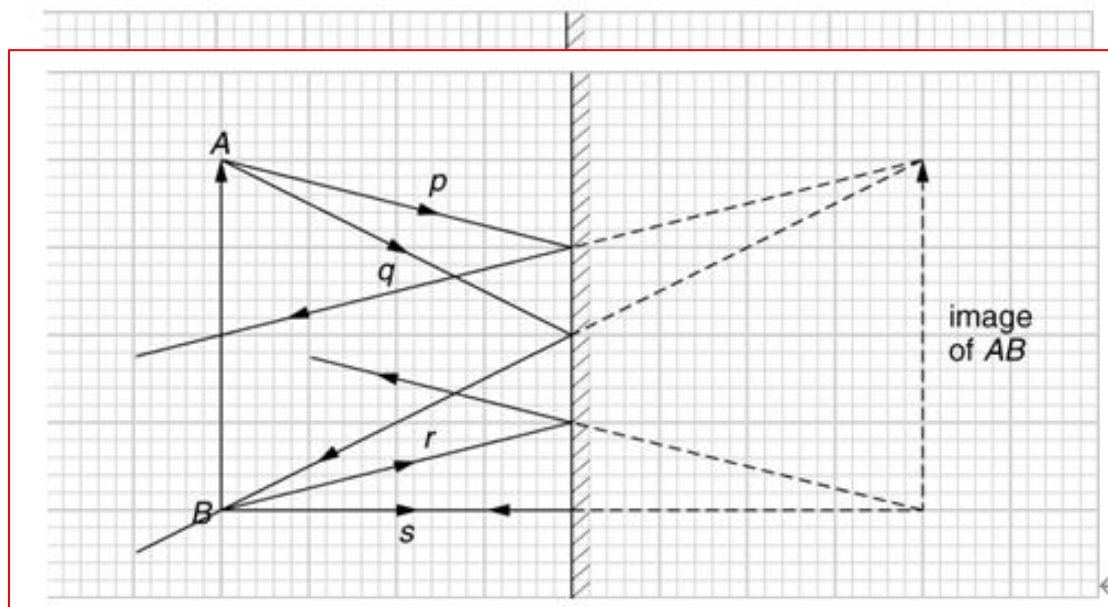
(a)



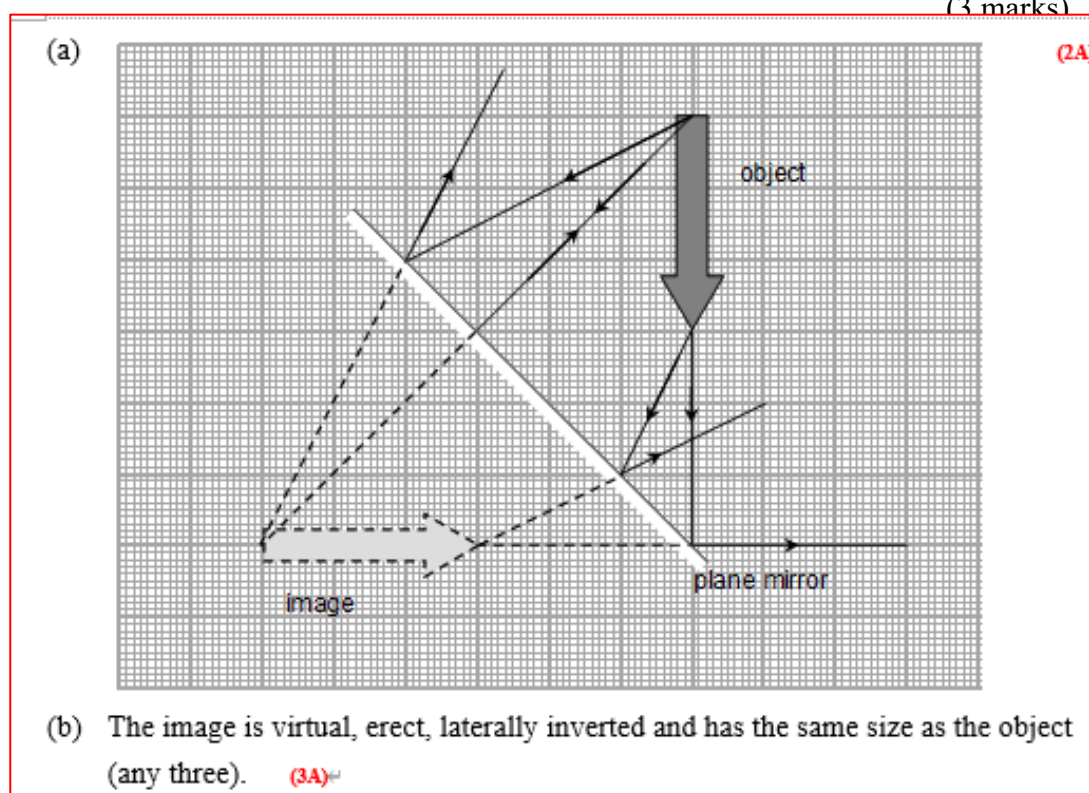
(b)



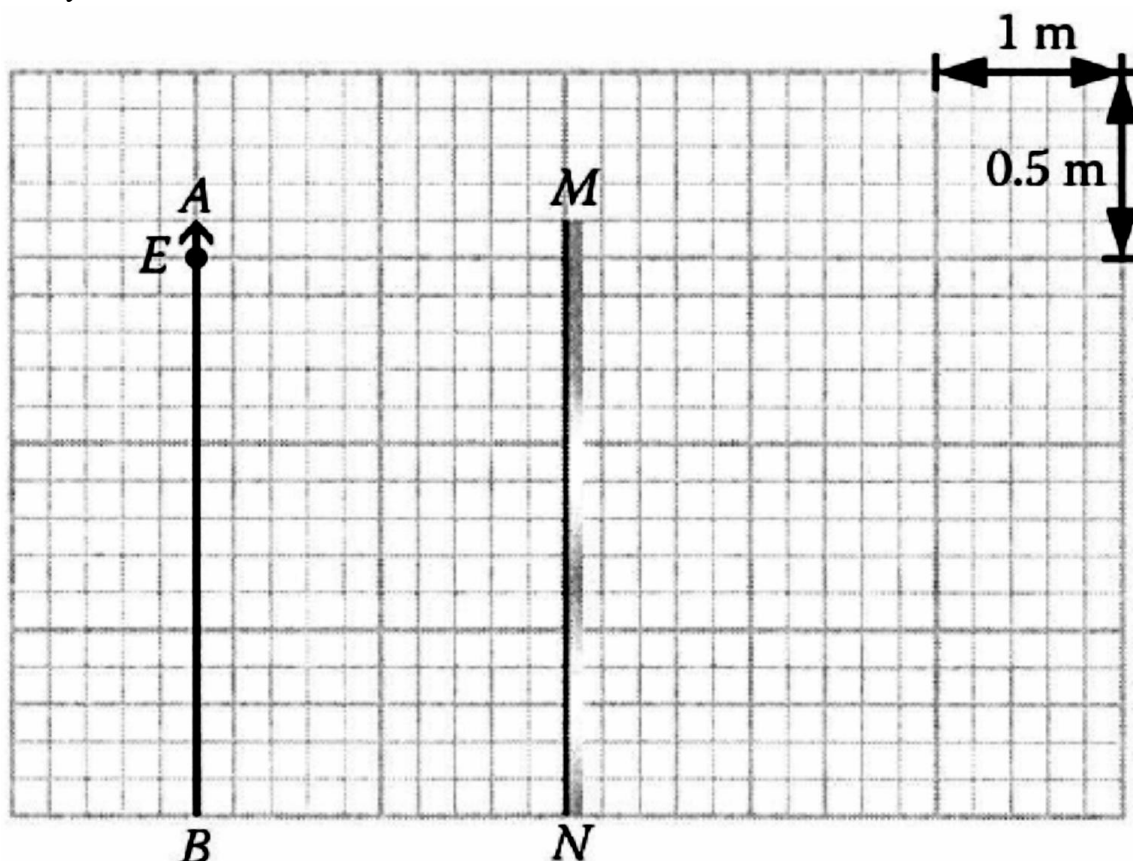
19. An object AB is placed in front of a plane mirror (Fig a).
 (i) Draw the reflected rays of p, q, r and s on Figure a. (4 marks)
 (ii) Hence mark on the figure the image of AB . (1 mark)



20. (a) An object is placed in front of a plane mirror as shown in the figure above. Copy the figure into a piece of graph paper, draw the reflected rays and mark the position of the image. (2 marks)
- (b) State three properties of the image formed. (3 marks)

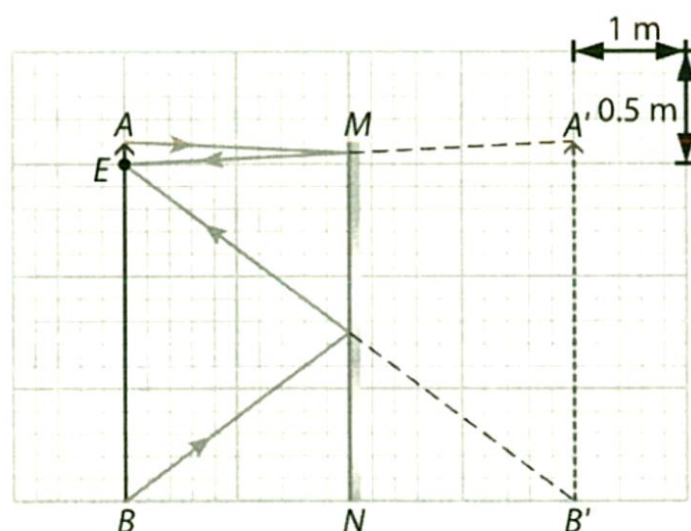


21. Sam (arrow AB) of height 160 cm is standing 2 m in front of a vertical plane mirror (MN). His eyes (E) are 150 cm above the ground.
- (a) (i) Draw, on the graph paper, a ray diagram to show how Sam sees his whole body in the mirror.
- (ii) Hence, or otherwise, find the minimum length of the mirror such that Sam can see his whole body.



(a) (i) Ray diagram:

(2A)



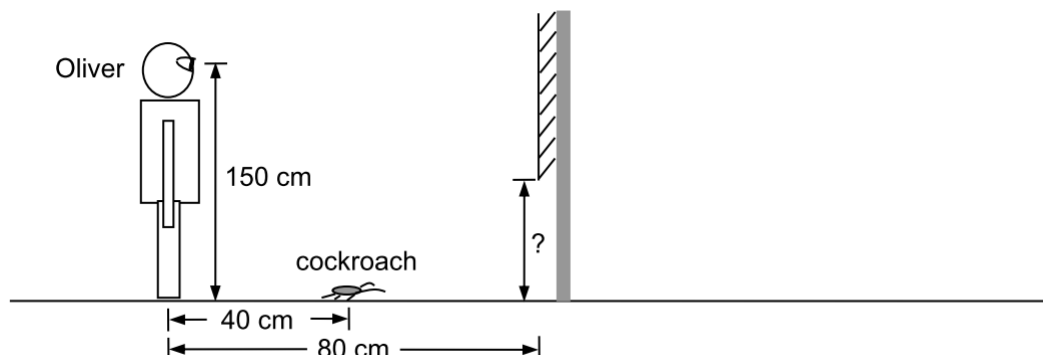
1A for correct rays

1A for correct image

(ii) Min. length = $(1.6)(0.5) = 0.8 \text{ m}$.

(1A)

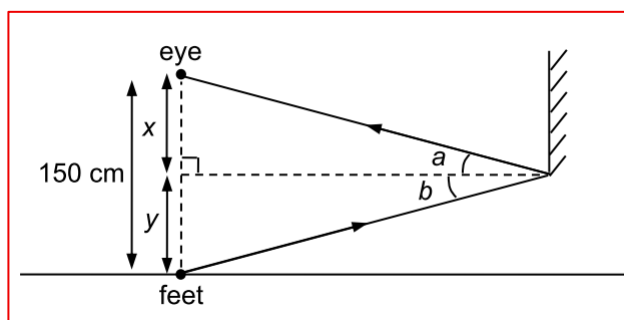
22. Oliver is 80 cm in front of a plane mirror. His eyes are 150 cm above the floor. There is a cockroach at 40 cm in front of him. How high above the floor is the bottom edge of the mirror if Oliver can just see (a) his feet, (b) the cockroach?



(a)

$$(a) \quad y = x = \frac{1}{2} \times 150 = 75$$

$\therefore$ The bottom edge of the mirror is 75 cm above the floor.

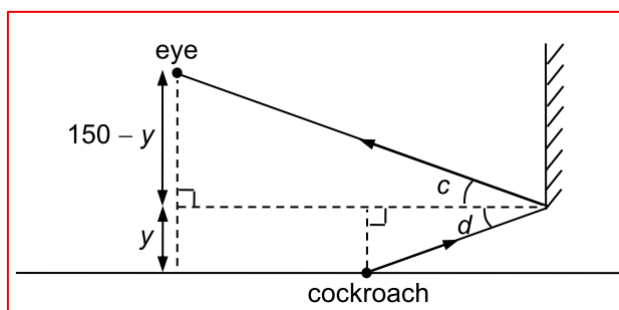


(b)

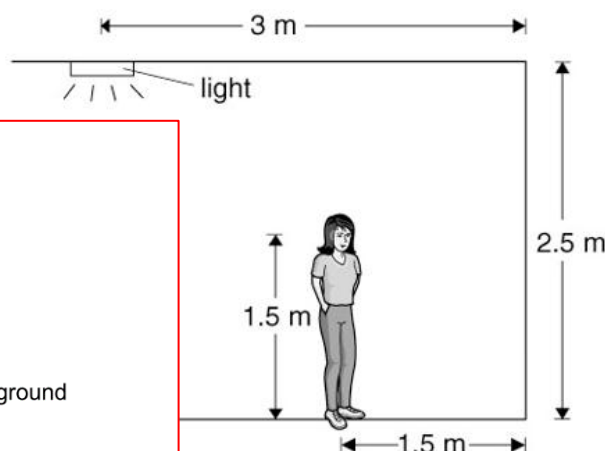
$$(b) \quad \frac{y}{150-y} = \frac{40}{80}$$

$$\Rightarrow y = 50$$

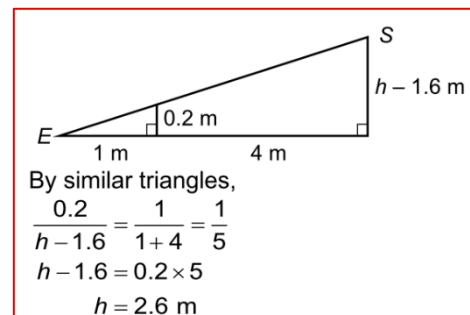
$\therefore$ The bottom edge of the mirror is 50 cm above the floor.



- 
- The diagram shows two right-angled triangles sharing a common vertex at point E. The larger triangle has a horizontal base of 3 m and a vertical height of 1 m, with the right angle at the bottom right. The smaller triangle has a horizontal base of 1.5 m and a vertical height of h, with the right angle at the bottom left. Both triangles are right-angled, and their corresponding sides are proportional, making them similar triangles.
- By similar triangles,
- $$\frac{h}{1.5} = \frac{1}{1.5+3}$$
- $$h = 0.33 \text{ m}$$
- Distance between the mirror and the ground
- $$= 1.5 + 0.33$$
- $$= 1.83 \text{ m}$$
- The mirror should be placed 1.83 m above the ground.

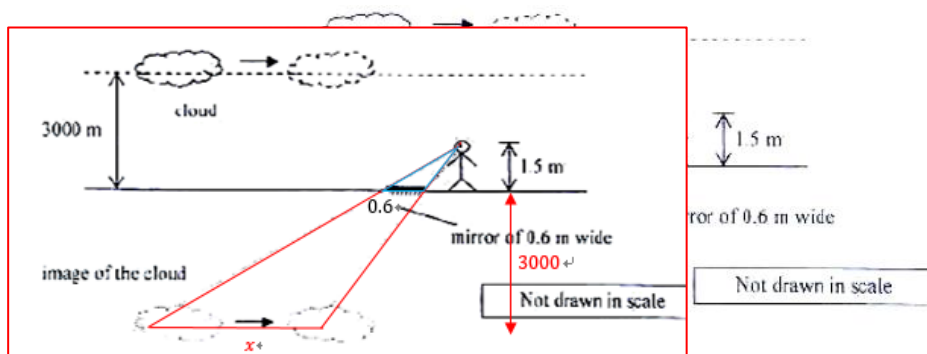


-
- Diagram for Question 10:
- A brick wall is on the left.
 - A spider is on the wall at height h .
 - John is standing in the middle, with a height of 1.6 m.
 - The distance from the wall to John is 3 m.
 - The distance from John to pole AB is 1 m.
 - Pole AB is on the right, with a height of 1.8 m.
 - A line of sight from the spider to the top of pole AB is shown, with a horizontal dashed line from John's eye level.
- Similar triangles are formed by the line of sight and the horizontal dashed line.
- By similar triangles:
- $$\frac{h - 1.6}{3} = \frac{1.8 - 1.6}{1}$$
- $$h - 1.6 = 0.2 \times 3$$
- $$h - 1.6 = 0.6$$
- $$h = 2.2$$

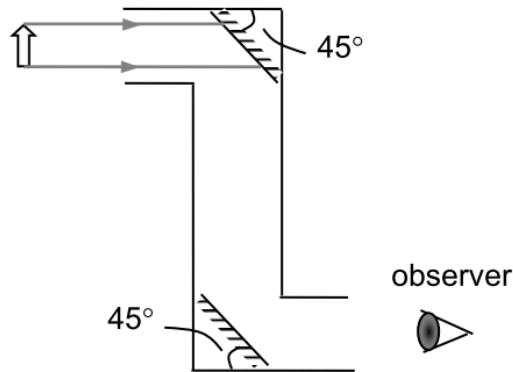


- What is the approximate speed of the cloud?

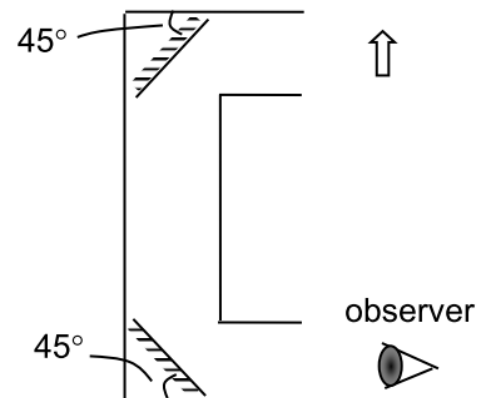
$$\begin{aligned}\frac{0.6}{x} &= \frac{1.5}{3000 + 1.5} \\ x &= 1200.6 \text{ m} \\ \text{Speed} &= \frac{d}{t} \\ &= \frac{1200.6}{20} \\ &= 60.03 \text{ ms}^{-1}\end{aligned}$$



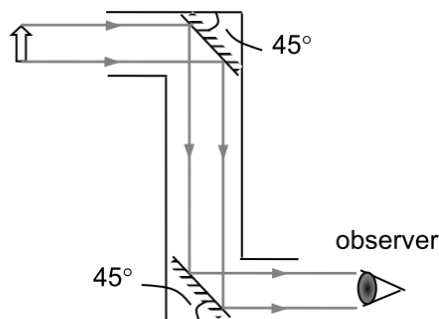
26. A periscope (潛望鏡) consists of two plane mirrors as shown. Each mirror makes an angle of 45° to the horizontal. Two rays of light from an object fall on the upper mirror in the periscope.



- Complete the paths of the rays until they reach the observer. (2 marks)
- Is the image seen by the observer erect or upside down? (1 mark)
- Is the image seen by the observer laterally inverted? (1 mark)
- The figure below shows another type of periscope.
 - Is the image seen by the observer erect or upside down? (1 mark)
 - Is the image seen by the observer laterally inverted? (1 mark)



(a)



(Each correct ray)

- Erect
- No
- (i) Upside down
(ii) No

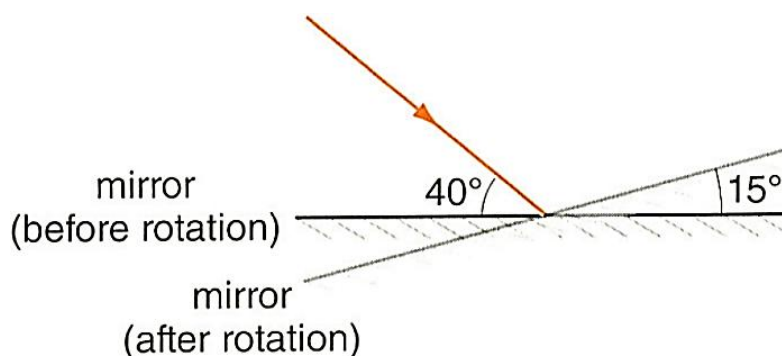
2 × 1A
1A
1A
1A
1A

27. (a) A plane mirror can be used as a rear-view driving mirror. State one advantage and one disadvantage of using the plane mirror as a driving mirror. (2 marks)
- (b) The Figure above shows an ambulance. Explain why the word AMBULANCE is printed in the form as shown in the figure. (2 marks)

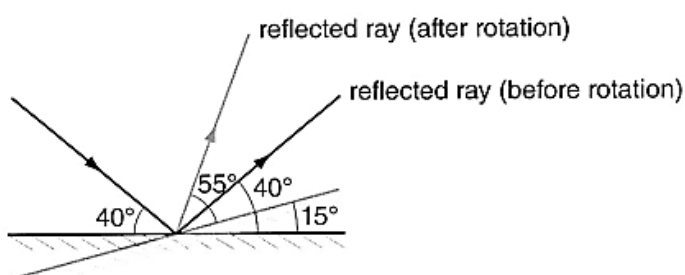


- (a) **Advantage** : the plane mirror can give same size images. [1]
- Disadvantage** : the plane mirror has multiple reflection that makes the image not clear. [1]
- (b) The drivers of cars in front can see the word "AMBULANCE" in the right way through the rear-view driving mirror. [1]

28. A light ray strikes a mirror as shown. The angle between the ray and the mirror is 40° . Then the mirror is rotated anticlockwise by 15° when the direction of the light ray remains unchanged. What is the angle by which the reflected ray turns?

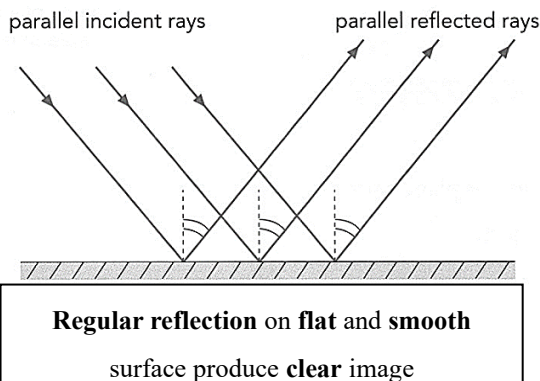


Angle between incident ray and mirror after rotation = $40^\circ + 15^\circ = 55^\circ$
The reflected rays are as shown.

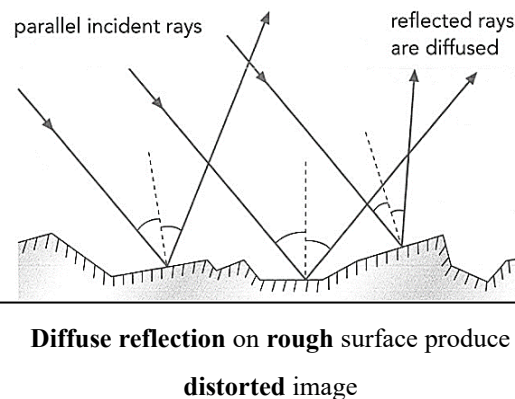


The required angle = $55^\circ - (40^\circ - 15^\circ) = 30^\circ$

Regular reflection



Diffuse reflection



29. Figure 8 shows a playground after raining. Images can be seen on the clam water surface of the wet ground.

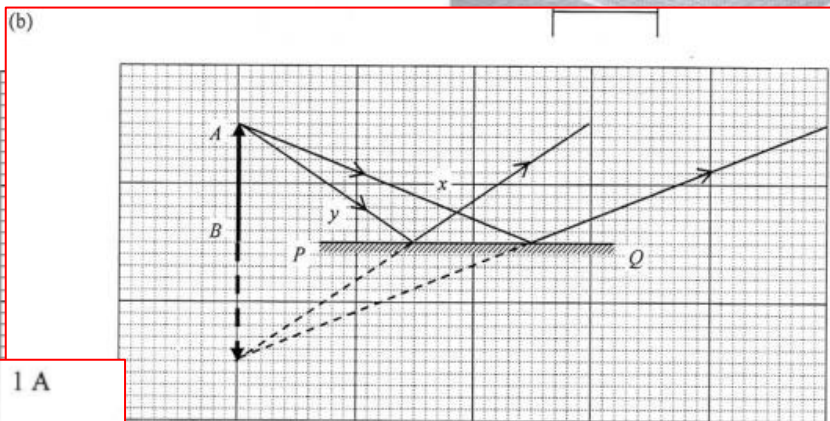
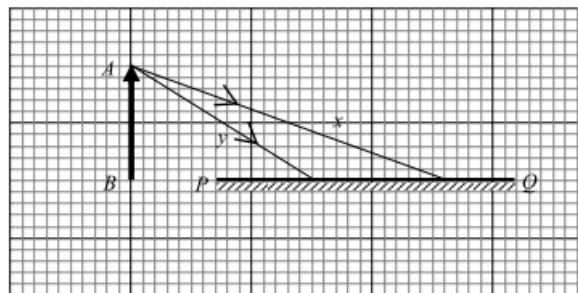
(a) Explain why images can be seen on the clam water surface.

(b) Figure 9 shows an object AB above the water surface PQ .

In figure 9,

(i) Draw the reflected rays of the incident rays x and y

(ii) Hence, draw the image of AB



(a) Calm water provides a smooth reflecting surface.

1 A

Regular reflection is occurred at the water surface.

1 A

30. The figures below are part of an advertisement for a face cream which was said to make people look younger by filling in any wrinkles in the skin. The manufacturer claimed that Figure (a) showed the effect of wrinkled skin on two rays of light and Figure (b) showed the effect after using the cream.

(a) Both figures contain errors of physics. Write a few words to point out the errors.

(b) Redraw the figures using the same incident rays and with the errors corrected.

(c) Name the types of reflection occurring in the figures.

(a) There should be only one reflected ray for each incident ray. Figure (a) violates this. (1)

Reflection of light obeys the law of reflection which states that the angle of reflection is equal to the angle of incidence. Figure (b) violates this.

(c) Diffuse reflection occurs in Figure (a) while regular reflection occurs in Figure (b). (1)+(1)

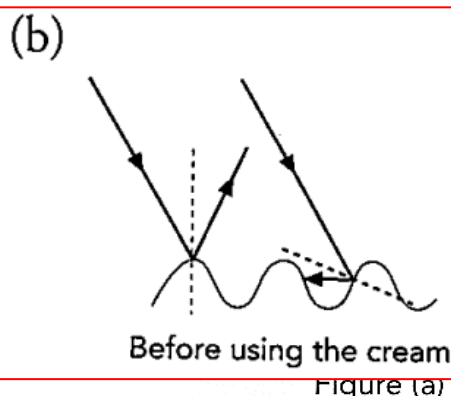
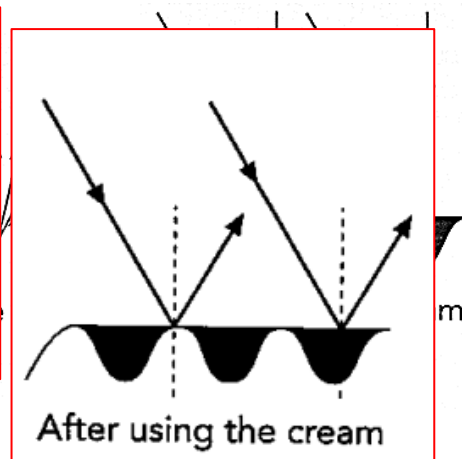


Figure (a)

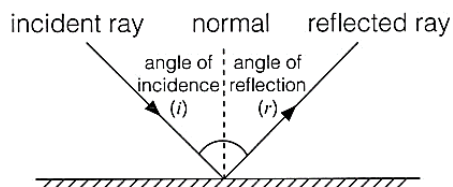


After using the cream

不可翻印,販賣,違者必究

31. Reflection of light

- (i) When light falls on an object, part of it is absorbed while the rest is _____.
- (ii) The reflection of light is governed by two laws, known as the laws of reflection.



The laws of reflection state that:

(1) _____

(2) _____

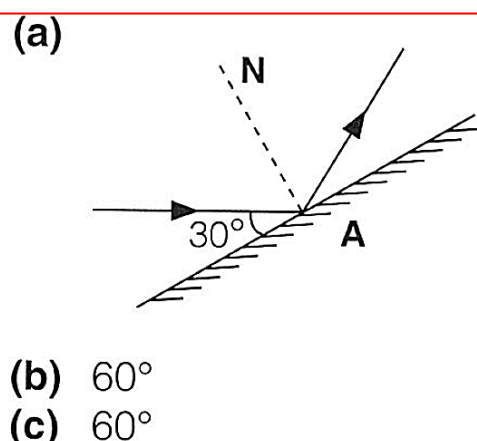
- (i) reflected
- (ii) (1) The incident ray, the normal and the reflected ray all lie on the same plane.
(2) The angle of incidence (i) is equal to the angle of reflection (r).

32. The diagram shows a light ray reflected by a mirror at point A.

(a) Draw a normal line at A and label it N.

(b) State the angle of incidence.

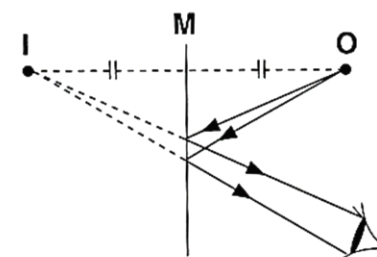
(c) State the angle of reflection.



33. Reflection on a plane mirror

The image formed in the mirror has the following properties:

- (a) The image is the same size as the object.
(b) The image is laterally inverted.
(c) The image is upright.
(d) The image is virtual.
(e) The image distance is same as the object distance.



(e) _____